

HDOC Day 21 - Answer

Algorithm

1. Start. 2. Input two positive integers m and n and ensure $m \leq n$. 3. Repeatedly display the menu. 4. Use switch-case for the selected operation. 5. For choices 1-4, use nested loops for every integer from m to n. 6. For an invalid choice, display an error message. 7. Repeat until choice 5 (Exit). 8. Stop.

C Program

```
#include <stdio.h>

int main() {
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n ( $m \leq n$ ): ");
        scanf("%d%d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Try again.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
        printf("\n1. First 10 multiples\n");
        printf("2. Addition tables\n");
        printf("3. Subtraction tables\n");
        printf("4. Division tables\n");
        printf("5. Exit\nEnter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                for (i = m; i <= n; i++) {
                    printf("\nMultiples of %d: ", i);
                    for (j = 1; j <= 10; j++) printf("%d ", i * j);
                    printf("\n");
                }
                break;
            case 2:
                for (i = m; i <= n; i++) {
                    printf("\nAddition table of %d\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d + %d = %d\n", i, j, i + j);
                }
                break;
            case 3:
                for (i = m; i <= n; i++) {
                    printf("\nSubtraction table of %d\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d - %d = %d\n", i, j, i - j);
                }
                break;
            case 4:
                for (i = m; i <= n; i++) {
                    printf("\nDivision table of %d\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d / %d = %.2f\n", i, j, (float)i / j);
                }
                break;
            case 5: printf("Exiting program...\n"); break;
            default: printf("Invalid menu choice.\n");
        }
    } while (choice != 5);
    return 0;
}
```

Test-case table

Test	Input	Choice	Expected result
1	m=2, n=3	1	Prints first 10 multiples of 2 and 3.

Test	Input	Choice	Expected result
2	m=2, n=2	2	Prints 2+1 through 2+10.
3	m=3, n=3	4	Prints 3/1 through 3/10 rounded to 2 decimal places.
4	m=5, n=2	-	Invalid input; asks again.
5	m=1, n=1	9	Invalid menu choice; menu continues.